The Future of Recruitment: Using Deep Learning to Build Intelligent Interview Bots

Shahnawaz Ahmad
School of Computer Science
Engineering and Technology
Bennett University
Greater Noida, India
shahnawaz98976@gmail.com

Shahadat Hussain
School of Computer Science
Engineering and Technology
Bennett University
Greater Noida, India
shahadat.hussain@bennett.edu.in

Mohammed Wasid
School of Computer Science
Engineering and Technology
Bennett University
Greater Noida, India
mohammed.wasid@bennett.edu.in

Ogala Justin Onyarin
Cyber Security Department
University of Delta
PMB 2090 Agbor,
Delta State, Nigeria
Justin.ogala@unidel.edu.ng

Mohd Arif
Department of Computer Science
Engineering and Technology
Galgotias University
Greater Noida, India
mohd.arif.jmi@gmail.com

Javed Ahmad
Department of Computer Science
Engineering and Technology
Sharda University
Greater Noida, India
javed.jmi08@gmail.com

2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10725310

Abstract*—The hiring process for IT positions is often complex and time-consuming, posing significant challenges for businesses seeking to maintain consistency and efficiency. Interview scheduling is complicated by manual effort and scheduling difficulties, which have been further made worse by the COVID-19 pandemic, making in-person interviews difficult. To address these challenges, this study proposes a Natural Language Processing (NLP)-based Machine Learning (ML) solution that automates interview scheduling and communication in natural language. To develop the solution, two deep-learning NLP models served as the foundation of the system, and AWS serverless stack and Amazon SageMaker were used for training and inference. The solution is designed to be scalable and cost-effective, and new models can be easily added and trained on unique subjects. Furthermore, the Amazon Sumerian scene is customizable to meet branding requirements. This research offers significant implications, as the proposed ML solution can help businesses streamline their hiring process, reduce human error, and improve the accuracy of candidate assessments. The scalability and cost-effectiveness of the solution make it ideal for businesses of all sizes. The smart interview bot developed in this study can conduct live-like interviews that mimic human interactions, making the hiring process more engaging and personalized. Overall, this paper demonstrates the potential of NLP to develop a smart solution for an everyday business problem. The intelligent live-like interview process and smart interview bot developed in this research have the potential to revolutionize the hiring process by automating scheduling and reducing manual effort. By leveraging the proposed ML solution, businesses can improve the accuracy and consistency of their hiring practices, ultimately leading to more successful outcomes.

Keywords: Machine Learning, Amazon Web Services, Natural Language Processing, BERT, Amazon SOS, Dynamo DB.

I. INTRODUCTION

Businesses find it tough to shortlist the top candidates since bringing consistency and efficiency to the interview process is challenging. It takes a lot of manual work to interview a candidate and assess his talents, and it's much more challenging to find the interviewer time to schedule and carry out the interviews. Given the COVID scenario, doing these interviews in person has become increasingly problematic, and a ML solution that can intelligently handle interview scheduling would surely benefit many organizations. A smart interview bot, also known as a chatbot, is an ML solution that can automatically schedule interviews and speak with humans in natural language [16].

Modelling conversation is a critical problem in natural language processing and artificial intelligence. Indeed, building a smart interview bot has remained one of AI's most complex issues since its inception. While the bot may be used for a variety of jobs, it must understand what users are saying and respond in a way that is relevant to the issue at hand. Deep learning has recently become more prevalent in businesses to shortlist the best candidates, and Chatbot is one of its applications [2, 3, 6, 7, 15]. It's sometimes difficult for businesses to shortlist the best candidates for IT jobs because it's challenging to bring consistency and efficiency to the way interviews are conducted. Interviewing an applicant and assessing his abilities takes a lot of manual effort, and finding the interviewer time to arrange and materialize the interviews is similarly tricky.

Given the COVID situation, doing these interviews in person has become increasingly complex. A ML solution that can intelligently manage interview scheduling would undoubtedly help many businesses improve their hiring process [12].

a. Challenges

The rise of remote work and the need for efficient recruitment processes have led to changes in the way job interviews are conducted. With most of the workforce working from home, the use of internet-based interviews has become increasingly common. However, the impact of this shift on the quality and effectiveness of the recruitment process is not yet fully understood. Therefore, it is important to investigate the potential benefits and limitations of using internet-based interviews and to develop new tools and techniques that can enhance the recruitment experience for both employers and candidates.

One of the most significant responsibilities of the hiring teams is to assess applicant profiles and get available interviewer and candidate slots to schedule the initial interview. As a result, following up with the interviewer to collect the interview comments and arrange the next steps takes time. Consider implementing this on a large scale with very high hiring objectives; there will be a lot of coordination necessary. Most businesses are searching for ways to use technology to provide applicants with better experience and

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.

speed up recruiting. The most critical challenges of the hiring process are.

- Working from home is the new norm.
- Scheduling interview.
- Time to revert with interview result.
- Conducting interviews at scale.
- Consistency in the candidate screening approach.

b. How to Address These Challenges?

To address the challenges, the solution proposed is to create an interactive interview experience using AWS services, where the candidate can record their responses, and a machine learning model is used to analyze them in near real-time [9]. This approach minimizes the need for interview availability and allows candidates to schedule interviews at their convenience [9]. The strategic steps used to address these challenges were:

- Interactive virtual interview user experience.
- Automate the evaluation of the candidate's subjective answer using machine learning.
- Near real-time evaluation at scale
- Bring in efficiency and consistency in the evaluation.
- Continuously improve the model inferencing accuracy

c. Key AWS Services Used

In this demo implementation, AWS services were employed to address the challenges of traditional interviewing processes, such as limited interview availability and inconsistent evaluation. An interactive interview experience was created in which an animated character played the role of the interviewer and had a dialogue with the candidate [10]. The candidate participated in this interactive discussion and recorded their responses. Machine learning was used to analyze candidate replies in near real-time, with a technique developed to increase the accuracy and speed of the machine learning model over time.

To implement this solution, a range of AWS services were utilized [9]. Amazon SageMaker was used to build, train, and deploy ML models for data persistence [1]. Amazon Dynamo DB, a no-sequel database that is easy to scale, was used for data storage. Amazon Lex was used to orchestrating the conversation, and Amazon Sumerian served as the VR engine for the frontend of the conversation. Furthermore, Amazon Simple Queue Service (Amazon SOS) and AWS Lambda were employed to decompose services for efficient implementation. The resulting solution enabled candidates to schedule interviews at their convenience, minimizing the need for interview availability and facilitating efficient and consistent evaluation.

d. Interview Evaluation Solution

To make asynchronous calls and isolate the service, we combined Amazon SQS and AWS Lambda. We utilized Amazon Lex to create a life-like dialogue; Sumerian violations and links can be integrated. The host interactions can be controlled to offer a compelling 3d, immersive experience. The following steps were taken for the interview evaluation processes.

- Amazon Sumerian host will interact with the user and capture interview answers.
- Amazon Sumerian will post the recorded answer to the Amazon Lex bot, invoking AWS Lambda functions.
- AWS to push the request to Amazon SOS.
- Amazon Lambda attached to the Amazon SOS will be triggered.
- Inference results will be updated back in Amazon Dynamo DB.

The architectural view of the interview evaluation is shown in figure 1.

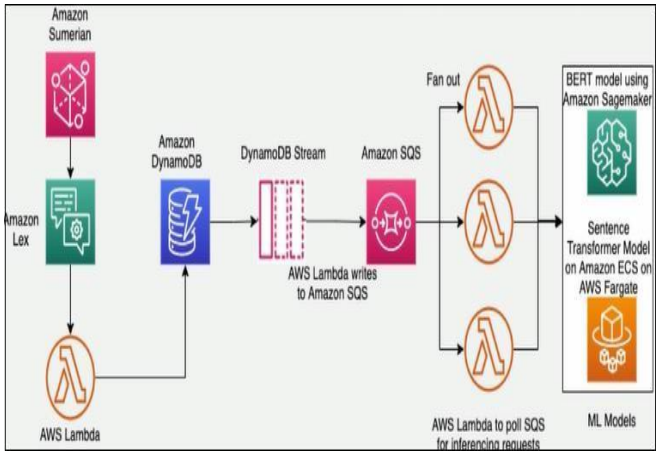


Fig. 1. Interview Evaluation Solution

The interview process involves the use of the Sumerian host to welcome the candidate and ask questions on the technical topic while capturing the candidate's response. Sumerian is configured with an Amazon Lex bot for the interview sessions, which can transcribe the audio. The Lex bot is set up with the appropriate intent to interact with the candidate. Once a question is asked, a Lambda function is called, which creates an entry in the Dynamo DB, recording the candidate's response.

To handle the ML inferencing, a BERT model has been launched as an Amazon SageMaker endpoint. Due to the unpredictable nature of the load on ML points and the maintenance required for the ML models, the solution has been designed for scalability and fault tolerance. The application has been decoupled from the inferencing task so that during peak usage, multiple Lambdas can be triggered to read the inferencing requests in batches and perform the inferencing synchronously. This approach ensures consistent performance of the application, even during high demand.

The Dynamo DB streams have been enabled to collect every new item published to Amazon SQS. The Machine Learning model is called in points for inferencing, and the results are sent back to the Dynamo DB using a Lambda that listens on the queue. Additionally, the Amazon mail service is leveraged to send an automated email to the candidate.

Overall, this solution addresses the challenges of scheduling and evaluating technical interviews in a scalable and efficient manner. The use of AWS services such as Amazon Sage Maker, Dynamo DB, Amazon Lex, and Amazon Sumerian enables the automation of the interview process while ensuring a high level of accuracy and consistency in the evaluation of candidates. The decoupling of the application from the inferencing task, along with the use of Dynamo DB streams and Amazon SQS, enhances scalability and fault tolerance.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.

II. LITERATURE REVIEW

There has been a surge in the development of smart interview bot systems in the business industry, such as retail, banking, and education. Researchers are working on making chatbots smarter and more like real people [4]. To achieve this, they are moving beyond simple rules and basic machine learning. Instead, they're using advanced techniques like NLP and deep learning, including deep neural networks (DNNs) and deep reinforcement learning (DRL). These fancy tools help chatbots better understand what you say and respond in a more natural way [13].

Traditionally, translation involved breaking sentences down into pieces and translating them one by one. Now, with neural machine translation, things work differently. This approach, inspired by deep learning and neural networks, treats the entire sentence. It's like a powerful model that can understand the entire conversation at once. This allows for more natural and accurate translations.

In tasks like translation and speech recognition, models need to handle sequences of information. A popular approach for this is the Sequence-to-Sequence model, which typically uses a type of neural network called a Vanilla RNN for both its "encoder" and "decoder" [15].

The encoder takes the source language sentence and tries to condense its meaning into a single "thought vector" - basically a bunch of numbers capturing the essence of the sentence. Then, the decoder uses this thought vector to generate the translated sentence in the target language.

However, there's a catch: Vanilla RNNs struggle when dealing with long sequences. The problem is that they must remember information from the beginning of the sequence all the way to the end, which becomes difficult for longer sentences. This can lead to inaccurate translations or errors in processing the information.

There are currently various performance measures for Virtual assistants, and specific measuring standards are followed throughout the industry [13]. Depending on the nature of their business and the market in which they operate, different organizations require Chatbots. The structure and length of a Chatbot's discussion are some of the most critical performance metrics. The size of the output sentence must be appropriate and relevant to the current topic. The faster they answer, the shorter and simpler the phrase structure in the output, and the higher the customer satisfaction rate. In banks, the Chatbot is primarily used to assist the user through the bank's policy, schemes, and other customer queries regarding their account to better grasp this measure. This allows users to complete activities faster and reduces the need for human call assistance, lowering service costs. Customer happiness also leads to a second indicator, the retention rate, critical to Chabot's performance.

To implement a demo of a smart interview bot system, this study utilized a variety of crucial AWS services [5]. At the centre of our solution are the machine learning models that enable us to automatically evaluate candidate answers. To build, train, and deploy these models, we used Amazon Sage Maker, which also provides a data persistence layer. Additionally, we utilized Amazon Dynamo DB, a no-sequel database that is easy to scale. And, for the front-end of the conversation, we used Amazon Sumerian as the VR engine.

During the interview, the Sumerian host greets the candidate and asks questions on our technical topic. It also captures the candidate's response to the question. Since the audio is the transcript of Amazon Lex, this is achieved by using Sumerian's ability to configure an Amazon Lex bot for the interview sessions [5]. Amazon Lex is configured with the correct intent to act on the candidate. To achieve the goal, the Lex bot will call a Lambda function. This Lambda function will create an entry in the DynamoDB with the candidate's response to the query. The enabled DynamoDB streams on the main DynamoDB table collect and push every new item published to Amazon SQS.

The solutions presented in the literature review are time-consuming, lack consistency and efficiency, and thus pose a significant challenge in business maintenance, especially during the COVID-19 pandemic. This work offers an NLP-based ML solution to automate interview scheduling and communication. We integrate deep-learning NLP models, the AWS serverless stack, and Amazon SageMaker to streamline the hiring process, making it scalable, cost-effective, and customizable. The aim is to develop a smart interview bot that reduces human error and enhances candidate assessment accuracy in the hiring process.

In the development of smart interview bot systems, various advanced concepts and techniques are being used, including NLP techniques and deep learning techniques such as DNN and DRL, in addition to traditional rule-based techniques and straightforward machine learning algorithms. NLP helps computers handle language in many ways, including translating languages, recognizing spoken words, and understanding written text.

There are various performance measures for virtual assistants, and specific measuring standards are followed throughout the industry. The structure and length of a Chatbot's discussion are critical performance metrics, with appropriate and relevant output sentence sizes, faster response times, and simpler phrase structures leading to higher customer satisfaction rates [14]. Banks use chatbots to help customers with policy questions, account inquiries, and other needs. This improves customer satisfaction and saves the bank money on service costs. However, it can be tricky to measure how well a chatbot is performing. Some indicators, like perplexity, learning rate, and BLEU score, can provide clues about how well the model is trained.

III. METHODOLOGY

The methods employed in this solution fall under the umbrella of NLP, a subfield of artificial intelligence that deals with the interaction between computers and humans using natural language. Both Bert and Sentence Transformer are NLP-based deep learning models that are designed to process and analyze human language. By leveraging these methods, the solution was able to achieve a higher level of accuracy and efficiency in evaluating candidate responses, ultimately resulting in better outcomes.

This study employed two deep learning models, namely Bert and Sentence Transformer, as the foundation of a solution aimed at achieving optimal scores. The primary model utilized was Bert, which is widely recognized as the most prevalent model for evaluating candidate responses. This

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.

model has been pre-trained with a vast amount of data and fine-tuned with training data, making it an ideal choice for the task at hand. Figure 2 illustrates the BERT-based model that served as the foundation of this solution.

In addition, the study incorporated a second model, namely Sentence Transformer, to further enhance the performance of the solution. This model is known for its ability to convert sentences into embedding, facilitating comparisons and analyses.

a. Model 1- BERT Based

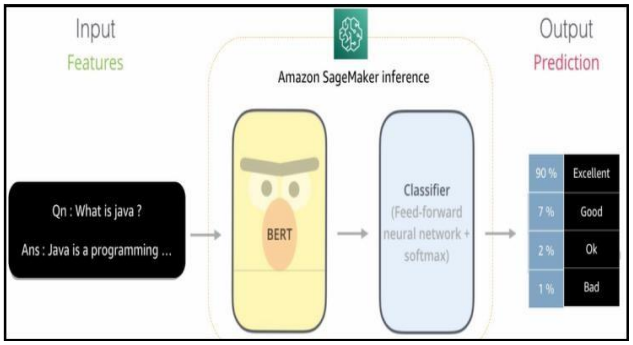


Fig. 2. – Model 1- BERT Based

The study utilized a specific machine learning model for each topic in evaluating candidates' responses in this section. While the Java model was trained, additional models can be created for other input aspects. Implementing separate models for each issue can enhance the evaluation of candidates' responses. The process involves transferring the question and response to a feedforward network for training, which rates the question and answers pairs as excellent, good, or bad. A training strategy was employed for the model, as shown in Figure 3, and will be briefly discussed.

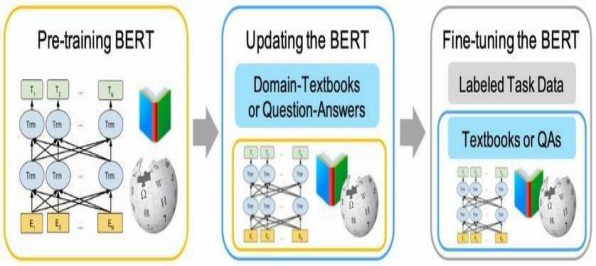


Fig. 3. Model 1- Retrained and fine-tuned BERT

In the second step, many labelled questions and answers were collected using Java, and the model has trained accordingly. The desired accuracy of 80% was achieved in just a few days, and the entire process of retraining and fine-tuning was carried out using TensorFlow with Amazon Sage Maker.

b. Model 2- Sentence Transformer Based

When using BERT for natural language understanding tasks, such as question answering or text classification, the model produces contextualized word embeddings for each token in the input text. To obtain a fixed-size representation of the entire input, pooling operations are often applied to the embeddings, such as max pooling or mean pooling.

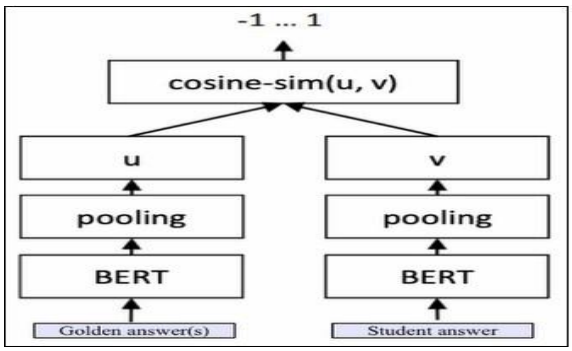


Fig. 4. Model 2 – Sentence Transformer Based

This results in a single vector that represents the entire input text. For evaluating a student's answer, the BERT model could produce contextualized embeddings for each token in both the gold answer and the student's answer. The embeddings for each answer could then be pooled to produce u and v, which would represent the fixed-size vector representations of the gold answer and the student's answer, respectively. The cosine similarity between these two vectors, as computed by the formula provided below, is used as a measure of similarity between the two answers.

The expression "cosine-sim(u,v)" represents the cosine similarity between two vectors u and v. Cosine similarity is a measure of similarity between two non-zero vectors that measures the cosine of the angle between them. It is often used in natural language processing and information retrieval to compare the similarity between two documents or sets of words.

The formula for cosine similarity between two vectors u and v is:

$$\text{cosine-sim}(u,v) = (u \cdot v) / (||u|| \, ||v||)$$

where "." denotes the dot product between the two vectors, and "||u||" and "||v||" denote the magnitudes or lengths of the two vectors.

To address the challenge of limited time and resources for creating training data, this study utilized an additional model alongside the BERT-based model. With this approach, new questions can be added and quickly assessed by simply providing a reference answer. During the assessment process, the candidate's response is compared to the reference answer, and a score is assigned based on the degree of similarity between the two. This model also utilizes a pre-trained BERT model for encoding and comparing the responses using cosine similarity. While this approach may not be as polite as the fine-tuned BERT model, it is effective for evaluating candidates on novel topics with limited training data. The entire system is built on AWS serverless services, including AWS Lambda, Amazon Dynamo DB, and Amazon SQS, with deep learning training and inference handled by Amazon Sage Maker.

IV. RESULTS AND ANALYSIS

The demo showcased a Java programming language interview, where the machine learning models were trained on the same topic. The interview was conducted using the Sumerian host, which asked the candidate two questions based on an MDP. The questions asked were: "What is the meaning of aggregation?" and "In Java, what is a transient variable?" The candidate's answers were evaluated to determine their performance. The Sumerian scene can be

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.

customized to meet branding needs, and the Lex bot can be programmed to ask more questions. It is also possible to train the machine-learning model on any technical topic to expand the scope beyond Java. After the interview, the candidate will receive the results on their registered DMA light.

In consideration of the demonstration, it is important to note that the solution can be customized to suit branding needs by personalizing the Sumerian and programming the Lex bot to ask additional questions. It is also noteworthy that the questions posed in the demonstration were related to the Java programming language. However, the system can be adapted for use in other technical fields by training the machine-learning model on the relevant subject matter.

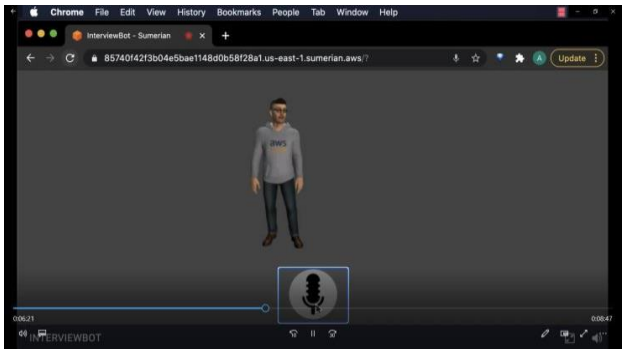


Fig. 5. Amazon Sumerian host interview demo

As depicted in Figure 6, the proposed solution involves three main stakeholders: the administrator, the candidate, and the evaluator. To ensure proper system architecture, it is essential to isolate the React-based front-end application from the AWS amplification components. The data is stored in both Amazon Dynamo DB and Amazon S3, and the API layer is managed by AWS AppSync. By separating the front-end from the back-end components, the solution can achieve better scalability, resilience, and security. Additionally, this approach allows for more efficient updates and maintenance of the system.

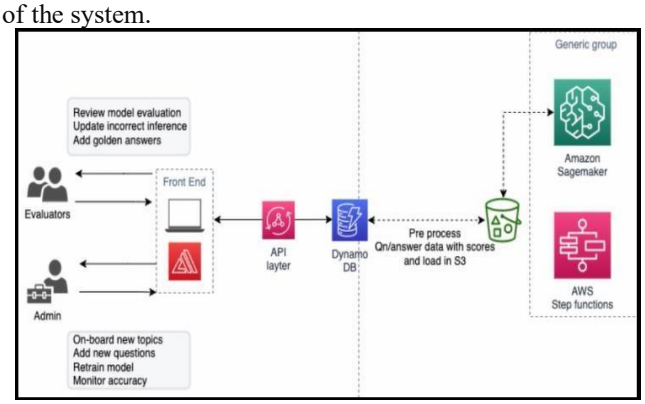


Fig. 6.- Admin/Evaluator Application

In addition, this system allows human evaluators to monitor automated evaluations and generate new training data to improve the model. The administrator can add new questions and refine existing ones to increase the model's accuracy. Ratings are determined based on the number of correct answers as evaluated by human raters compared to the total number of answers.

V. DISCUSSION

In this study, an intelligent interview bot was developed using deep learning with AWS services, and two deep learning models were employed to ensure the best possible score. The first model was a fine-tuned BERT-based model, while the second model was a sentence, transformer-based model. Both models were used to train the bot to ask and evaluate candidate responses to technical questions in the Java programming language. The results of the study showed that the intelligent interview bot was effective in evaluating candidates based on their responses to the technical questions asked. The system was built on the AWS serverless stack and training/inference using Amazon SageMaker, which allowed for quick scaling and cost control. Moreover, the system was easy to customize, with the Amazon Sumerian scene being customizable to meet branding needs, and new models could be quickly added and trained on unique subjects.

One limitation of this study is that the bot was trained on technical questions in the Java programming language only. However, as discussed in the results section, the system can be expanded to include other technical areas by training the machine-learning model on any technical topic.

In conclusion, the intelligent interview bot developed in this study has the potential to revolutionize the recruitment process, making it more efficient and objective. The system's ability to evaluate candidates based on their responses to technical questions could be extended to include other industries, such as finance and healthcare, and customized to fit the requirements of different companies. Further research can be conducted to improve the bot's performance and to explore its potential in other industries.

VI. CONCLUSION

Overall, this paper presents a valuable contribution to the development of an intelligent interview bot using deep learning and AWS services. The proposed system offers a scalable and cost-effective solution for interview scheduling and evaluation. The use of deep learning models has ensured the best possible score, and the AWS serverless stack has enabled quick scalability and cost control.

However, some limitations of this study should be acknowledged. First, the proposed solution has only been tested on a limited number of participants, and further research may be needed to evaluate its performance on a larger scale. Second, the study has not addressed the potential ethical concerns associated with using an intelligent interview bot in the hiring process, such as biases in the model or privacy issues. These limitations suggest avenues for future research to explore the full potential and implications of using an intelligent interview bot in the recruitment process.

In conclusion, this study provides a promising approach to addressing the challenges of the recruitment process through the development of an intelligent interview bot using deep learning and AWS services. However, further research is needed to fully understand the potential benefits and limitations of this technology, as well as to address the ethical considerations associated with its implementation.

VII. LIMITATIONS AND RECOMMENDATION

The proposed solution of creating an interactive interview experience using machine learning models also has some limitations that need to be acknowledged. These limitations include:

- Dependence on training data quality and diversity: The accuracy of the machine learning model used to

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.

evaluate the candidate's responses is dependent on the quality and diversity of the data used to train it. If the training data is biased or limited, the model may not be able to accurately evaluate all candidate responses.

- Limited suitability for certain job roles and industries: The use of an interactive virtual interview experience may not be suitable for all types of job roles or industries. For example, roles that require physical presence or hands-on skills may not be suitable for this approach.
- Candidate discomfort with technology: Some candidates may not be comfortable with the technology and may prefer more traditional interview methods. It is essential to ensure that the candidate's experience is positive and inclusive, and that the technology does not hinder or disadvantage any applicants.
- An initial investment of time and resources: Implementing a new technology solution like this requires an initial investment of time and resources, which may not be feasible for all organizations, particularly small businesses with limited budgets. Therefore, it is essential to assess the cost and benefits of implementing such a solution before deciding to adopt it.

To fully realize the potential of this solution, here are some recommendations based on the research findings:

- Improve the quality and diversity of the training data used to train the machine learning models to ensure better accuracy in evaluating candidate responses.
- Consider the suitability of an interactive virtual interview experience for different types of job roles and industries before implementing the solution.
- Ensure the candidate's experience is positive and inclusive and that the technology does not hinder or disadvantage any applicants.
- Provide training and support for candidates who may not be comfortable with the technology to ensure equal opportunities for all candidates.
- Evaluate the cost and benefits of implementing a new technology solution like this before deciding to adopt it.
- Continuously monitor and evaluate the effectiveness of the solution and make necessary adjustments to improve its performance.
- Explore the possibility of integrating additional features, such as sentiment analysis or natural language processing, to enhance the machine learning models' performance.

REFERENCE

[1]. Amazon Web Services, Inc. n.d. Machine Learning – Amazon Web Services. https://aws.amazon.com/sagemaker/.

[2]. Beaudry, A., Boucher, S., Landry, R. (2019). Deep learning applications in HR research: Emerging trends and implications for practice. Human Resource Management Review, 29(1), 98–113.

[3]. Beaudry, J., Consigli, A., Clark, C., Robinson, K. J. (2019). Getting ready for adult healthcare: Designing a Chatbot to coach adolescents with special health needs through care transitions. Journal of Pediatric Nursing, 49, 85– 91. https://doi.org/10.1016/j.pedn.2019.09.004

[4]. Chakraborty, D., Ray, D., Mukherjee, A., & Naskar, R. (2020). Chatbot development using machine learning techniques: A review. In Proceedings of the 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE.

[5]. Chen, M., Wang, L., Li, Y., & Lin, Z. (2017). A demo of a smart interview bot: Building a real-time VR interview system using Amazon Sumerian. In Proceedings of the 2017 ACM Conference on Information and Knowledge Management (pp. 2497-2499).

[6]. Heller, B., Proctor, M., Mah, D., Jewell, L., & Cheung, B. (2005). Freudbot: An investigation of Chatbot technology in distance education. In EdMedia+ Innovate Learning, pp. 3913-3918. Association for the Advancement of Computing in Education (AACE).

[7]. Heller, J., Bhattacharyya, S., & Reed, J. (2005). Learning to converse with a spoken dialogue system in a new domain. Computer Speech & Language, 19(2), 137– 158.

[8]. Improving Service and Reducing Costs in Contact Centers How adding intelligence optimizes your business and customer experience. https://hqn.thinkmethodology.co.uk/wpcontent/uploads/2021/11/Improving-service-andreducing-costs-in-contact-centres-AWS.pdf.

[9]. Jafari, M. (2021). An interactive virtual interview experience using AWS. Medium. Retrieved from https://towardsdatascience.com/an-interactivevirtual-interview-experience-using-aws-ee53ab72c299

[10]. Chia-Wei Liu, Ryan Lowe, Iulian Serban, Mike Noseworthy, Laurent Charlin, and Joelle Pineau. 2016. How NOT To Evaluate Your Dialogue System: An Empirical Study of Unsupervised Evaluation Metrics for Dialogue Response Generation. In Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, pages 2122–2132, Austin, Texas. Association for Computational Linguistics. Copy Markdown to ClipboardCopy ACL to Clipboard.

[11]. The Machine Learning Journey The path toward leveraging the full power of machine learning technologies. https://pages.awscloud.com/rs/112-TZM-766/images/the-machine-learning-journey.pdf

[12]. Przegalinska, A., Ciechanowski, L., Stroz, A., Gloor, P., & Mazurek, G. (2019). Ciechanowski. L, Stroz. A, Gloor. P, and Mazurek. G (2019). In bot, we trust: A new methodology of Chatbot performance measures. Business Horizons, 62(6), 785–797. https://doi.org/10.1016/j.bushor.2019.08.005

[13]. 7 Leading Machine Learning Use Cases. https://resources.aithority.com/ait-whitepapers/7leading-machine-learning-use-cases.pdf. [Accessed 1 March 2022].

[14]. Alex Sherstinsky, Fundamentals of Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) network, Physica D: Nonlinear Phenomena, Volume 404, 2020, 132306, ISSN 0167-2789, https://doi.org/10.1016/j.physd.2019.132306.

[15]. Sutoyo, R., Chowanda, A., Kurniati, A., Wongso, R. (2019). Designing an emotionally realistic chatbot framework to enhance its believability with AIML and information states. Procedia Computer Science, 157, 621–628. https://doi.org/10.1016/j.procs.2019.08.226

[16]. Sutskever, I., Vinyals, O., & Quoc, V. (2014). Le. Sequence to sequence learning with neural networks. In Advances in neural information processing systems, pp. 3104-3112. 2014.

[17]. Narkhede, V., Surushe, O., Kulkarni, S., Solanki, H., Ekbote, T., & Joshi, D. (2023, March). AVA: A Photorealistic AI Bot for Human-like Interaction and Extended Reality. In 2023 International Conference on Communication System, Computing and IT Applications (CSCITA) (pp. 5-9). IEEE

[18]. Jadhav, B., Sawant, A., Shah, A., Vemula, P., Waikar, A., & Yadav, S. (2024, March). A Comprehensive Study and Implementation of the Mock Interview Simulator with AI and Pose-Based Interaction. In 2024 1st International Conference on Cognitive, Green and Ubiquitous Computing (IC-CGU) (pp. 01-05). IEEE.

[19]. Shahnawaz Ahmad, Iman Shakeel, Shabana Mehfuz, Javed Ahmad, Deep learning models for cloud, edge, fog, and IoT computing paradigms: Survey, recent advances, and future directions, Computer Science Review, Volume 49, 2023, 100568, ISSN 1574-0137, https://doi.org/10.1016/j.cosrev.2023.100568.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 07,2024 at 07:52:11 UTC from IEEE Xplore. Restrictions apply.